

Machine Learning in Applications

Group 21 - Project Report

A comparative analysis of the performance of different generative methods for virtual staining for histopathology

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Abstract—Primary Aldosteronism (PA) is the leading cause of secondary hypertension, and its treatment requires precise localization of aldosterone-producing regions via CYP11B2 immunohistochemistry (IHC). IHC, however, is expensive and resource-intensive, whereas Hematoxylin and Eosin (H&E) staining is cheap and ubiquitous but does not directly express molecular signals. We present a deep learning framework for *virtual staining* that predicts CYP11B2 expression from H&E images, comparing four architectures: two GAN-based methods (MCCL, UNISTainNet), a latent diffusion framework (HistDiST), and SwinNet, our deterministic dual-branch model combining a pathology-pretrained CTransPath encoder with a structure-preserving U-Net decoder. We evaluate all models at pixel level against ground-truth IHC and, *crucially*, at the structural level via segmentation of aldosterone-producing regions — assessing downstream diagnostic utility rather than visual realism alone. Experiments on a proprietary dataset from two patients reveal that translating generic tissue morphology into sparse molecular markers remains highly challenging: diffusion models are severely penalized by H&E/IHC spatial misalignment, while adversarial and deterministic approaches oscillate between false-positive hallucination and overly conservative predictions. UNISTainNet shows partial success in localizing densely positive regions, but no architecture reliably synthesizes sparse IHC targets from unaligned data, pointing to the need for more explicit spatial alignment and conditioning mechanisms.

I. INTRODUCTION

Primary Aldosteronism (PA) represents the most common etiology of secondary hypertension. Accurate patient stratification and subsequent surgical or medical interventions depend heavily on the identification and spatial characterization of aldosterone-producing regions. Clinically, this is achieved via CYP11B2 immunohistochemistry (IHC), which enables pathologists to localize functional nodules and apply the HISTALDO classification system. Despite its diagnostic value, IHC is fundamentally limited by high costs, tissue consumption, and prolonged turnaround times, rendering it unsuitable for ubiquitous adoption, particularly in resource-constrained settings.

Conversely, Hematoxylin and Eosin (H&E) staining is the cornerstone of standard histopathology—widely accessible, rapid, and inexpensive. However, H&E relies purely on morphological features and does not natively express the molecular signals captured by IHC. Recent advancements in deep learning and computational pathology have demonstrated that morphological patterns in H&E slides often harbor latent representations of molecular phenomena. This intersection presents a compelling opportunity: leveraging generative Artificial Intelligence to infer molecular profiles directly from routine H&E WSIs.

This study introduces a comprehensive deep learning framework designed to synthesize virtual CYP11B2 IHC representations from paired H&E WSIs. We specifically investigate the efficacy of four distinct, cutting-edge architectures — Multi Channel Correlation Learning (MCCL) [1] and UNISTainNet [2], both adversarial-based methods; SwinNet, a deterministic dual-branch architecture based on a pathology-pretrained Swin/CTransPath semantic encoder and a full-resolution structural branch; and a latent diffusion framework (HistDiST) [3] — to approximate the complex spatial distribution of aldosterone-producing regions.

To ensure clinical relevance, we propose a two-tiered validation strategy. First, we perform a direct differential comparison using image-level metrics to evaluate staining fidelity. Second, and *crucially*, we evaluate the models based on downstream task performance by segmenting the regions of interest (ROIs). By comparing the aldosterone-producing regions segmented on the virtual IHC against the corresponding CYP11B2-positive regions extracted from the real IHC, we assess the preservation of critical diagnostic features such as nodule localization, extent, and morphology. The entire framework is trained and evaluated on a highly constrained proprietary clinical dataset comprising biopsies from two distinct patients. This approach provides critical insights into the feasibility and robustness of virtual staining under severe data scarcity in real-world PA diagnostics.

II. BACKGROUND

A. Virtual Staining in Computational Pathology

The translation of standard H&E images into specialized stains has gained significant traction. Early approaches relied heavily on Convolutional Neural Networks (CNNs) and standard Generative Adversarial Networks (GANs), such as Pix2Pix [4] and CycleGAN [5], to perform image-to-image translation. While effective for simple color mapping, these early formulations struggle with complex, structurally dependent molecular targets and are prone to hallucinatory artifacts. Subsequent advancements aimed to enforce stronger morphological consistency.

B. Pathology Foundation Models for Conditioning

A pivotal breakthrough in recent virtual staining frameworks is the integration of pathology-specific foundation models, such as UNI [6], Phikon [7] and GigaPath [8]. Trained via self-supervised learning on massive cohorts of highly diverse Whole Slide Images (WSIs), these models extract robust, scale-aware feature embeddings that capture intricate tissue

morphologies. Rather than relying solely on the generative network to learn cellular structures from scratch, state-of-the-art architectures leverage these foundation models to heavily condition the generation process. Indeed, the recent leap in structural fidelity and generalization—shared across diverse architectures including MCCL, UNIStainNet, and HistDiST—stems directly from utilizing these pre-trained visual encoders. By injecting foundation-level embeddings into the generative pipeline, these networks can better contextualize dense cellular neighborhoods, accurately mapping H&E morphology to precise immunohistochemical signals.

C. GAN-based Virtual Staining

Generative Adversarial Networks remain one of the most widely adopted paradigms for H&E-to-IHC translation, owing to their ability to synthesize sharp, realistic textures in a single forward pass. The central difficulty is that paired H&E and IHC images originate from consecutive tissue sections and are therefore never perfectly aligned, making naive pixel-level supervision unreliable. Modern GAN-based methods address this by relaxing strict pixel correspondence — through contrastive patch-level objectives, similarity-based down-weighting of inconsistent regions, and pathology-aware constraints that preserve molecular semantics rather than exact geometry — and, increasingly, by injecting spatial guidance directly into the generator via spatially-adaptive normalization (SPADE) [9]. In this work we evaluate two GAN-based architectures, MCCL and UNIStainNet, which instantiate complementary versions of these ideas.

Multi Channel Correlation Learning (MCCL) [1] is a virtual staining framework optimized to translate Hematoxylin and Eosin (H&E) tissue representations into synthetic Immunohistochemistry (IHC) modalities. Unlike standard RGB architectures, MCCL embeds a fully differentiable color deconvolution layer directly into the optimization pipeline to unbundle raw pixels into distinct biological optical density channels, isolating target expressions like Diaminobenzidine (DAB) brown protein. To preserve complex tissue architecture and high-level biological semantics, the framework leverages the deep histopathological embeddings of Prov-GigaPath, a frozen pathology foundation model.

UNIStainNet [2] is a SPADE-UNet whose generator is conditioned on *dense* spatial tokens extracted from the frozen UNI [6] pathology foundation model, providing tissue-level semantic guidance at every decoder stage. Rather than engineering the supervision signal, it tolerates consecutive-section misalignment through a dedicated, misalignment-aware loss suite — reduced-resolution perceptual losses, an unconditional discriminator, and a DAB intensity term for stain quantification — while still retaining paired training. We describe its architecture in detail in Section III-D2.

D. Transformer-Based and Stain-Aware Architectures

Transformer-based encoders have recently emerged as an alternative to purely convolutional feature extractors in computational pathology. By modeling long-range dependencies and multiscale tissue context, Swin Transformer-based models

can complement the fine local information captured by CNNs. CTransPath [10], which combines convolutional inductive biases with a hierarchical Swin Transformer and pathology-specific pre-training, has shown that these representations can effectively encode both cellular morphology and broader tissue organization.

For virtual staining, combining semantic transformer features with high-resolution structural representations offers a way to preserve local morphology while estimating stain-specific molecular patterns. This is especially relevant for CYP11B2-IHC synthesis, where DAB-positive regions are sparse and their spatial localization is more informative than global color similarity alone. These principles motivate structure- and stain-aware architectures such as SwinNet, which complement adversarial and diffusion-based approaches through deterministic paired translation.

E. Diffusion Models for High-Fidelity Generation

Denoising Diffusion Probabilistic Models (DDPMs) [11] and Denoising Diffusion Implicit Models (DDIM) [12] have emerged as the state-of-the-art in generative AI, offering superior training stability and image quality compared to traditional GANs. In digital pathology, models combining diffusion with transformer architectures, such as HistDiT [13], have demonstrated the potential of framing virtual staining as a conditional generation process to synthesize biologically accurate molecular representations.

To practically realize this paradigm, our methodology employs HistDiT [3], an advanced Latent Diffusion Model (LDM) tailored for high-fidelity H&E-to-IHC translation. HistDiT addresses the inherent limitations of standard diffusion models in medical imaging—such as brightness biases and structural degradation—by integrating optimized noise schedules and specialized inversion techniques during inference. Coupled with the foundation-level conditioning discussed above (Phikon, in this case), HistDiT strikes an optimal balance between strict morphological preservation and the generative flexibility needed to capture complex IHC expressions.

Part of our work builds upon this framework, evaluating its capability against advanced GANs to precisely reconstruct the localized, hormone-producing microstructures critical for PA diagnosis.

III. MATERIALS AND METHODS

A. Dataset

The dataset utilized in this study comprises a highly controlled cohort of 13 paired Whole Slide Images (WSIs), featuring standard Hematoxylin and Eosin (H&E) staining and their corresponding CYP11B2 immunohistochemistry. Due to the limited availability of patient biopsies and the necessity to rigorously evaluate the models’ out-of-distribution (OOD) generalization capabilities, the dataset was partitioned strictly at the patient level.

The clinical cohort includes tissue samples from two distinct patients. Data from the first patient (58 years old), consisting of 10 WSIs, was utilized for model development. This subset was further divided into a training set of 8 WSIs—which,

upon processing, yielded a comprehensive training corpus of approximately 17,000 patches—and a validation set of 2 WSIs ($\approx 4,000$ patches). The validation set was employed exclusively for best model selection and qualitative visual validation. Because standard optimization loss metrics often correlate poorly with perceptual quality, structural fidelity, and clinical plausibility in generative models, visual inspection of the generated virtual stains was prioritized to identify the best-performing checkpoints. This approach ensures that the final evaluation on the test set accurately reflects true generalization performance rather than merely an optimized mathematical objective. To effectively benchmark the robustness of the virtual staining frameworks against real-world domain shifts, the remaining 3 WSIs ($\approx 4,500$ patches), sourced from the second patient (70 years old), were held out as an independent testing set. Evaluating the generated virtual IHC on a completely unseen patient provides a crucial and stringent measure of the models’ clinical viability, structural fidelity, and ability to generalize across different biological profiles.

Beyond the restricted patient cohort, an additional significant challenge inherent to this dataset is the severe structural imbalance between CYP11B2-positive (aldosterone-producing) regions and the surrounding negative tissue. Specifically, the ratio of patches containing functional, hormone-producing nodules (characterized by brown chromogen deposition on the ground-truth IHC) to non-producing patches is approximately 1:6 ($\approx 3,900$ positive out of $\approx 25,400$ total patches). Left unaddressed, this disproportion heavily biases generative models toward synthesizing predominantly negative tissue, thereby degrading the diagnostic utility of the virtual stain. To counteract this, data augmentation techniques were applied during the training phase. This strategy ensured a more balanced representation, forcing the networks to robustly learn and reconstruct the complex morphological signatures associated with aldosterone production.

B. Preprocessing

The raw data consist of paired H&E and CYP11B2 IHC Whole Slide Images stored as gigapixel .ndpi files, which we convert into a corpus of aligned and tissue-rich patches. After pairing each H&E slide with its IHC counterpart, we register the pairs with the VALIS framework [14]: a *rigid* step corrects the bulk offset and a *non-rigid* step compensates local deformations. Measured on our slides, this reduces the mean registration distance from up to ≈ 1.3 cm to roughly $100 \mu\text{m}$, and the relative Target Registration Error (rTRE) — which measures how far corresponding tissue landmarks still lie apart, relative to the image size — by one to two orders of magnitude (from up to 48.6 down to 0.33–0.44). Registration cannot, however, eliminate the intrinsic differences between two physically distinct sections — the residual misalignment reported for such consecutive-section pairs is on the order of 10–50 px [2] — which we account for explicitly in the model and loss design (Section III-D2).

From the aligned slides we extract spatially-corresponding 1100×1100 px patches at full magnification (stride 1024 px), discarding background and low-information tiles with a combined HSV + variance criterion (grayscale variance > 150 ,

HSV tissue ratio > 0.8). Before being fed to any model, patches are reduced to 512×512 px: during training via a random 1024×1024 crop followed by bicubic downscaling, and at evaluation via a centre crop of the same size followed by the same downscaling. Each IHC patch is then labelled *stained* or *unstained* according to its fraction of brown DAB pixels in HSV space, and finally exported as high-quality JPEG (quality 95) — which substantially reduces the on-disk footprint of the tens of thousands of 1100×1100 patches at negligible visual loss — together with a per-patch binary DAB mask used as ground truth for the region-level evaluation.

All thresholds were calibrated through an Exploratory Data Analysis rather than chosen arbitrarily. We compared three tissue-pruning strategies — a brightness-based *Basic* filter, an *Otsu* grayscale detector, and the combined HSV + variance criterion — and, for each, swept its threshold while monitoring the patch-retention rate. HSV + variance (var > 150 , tissue ratio > 0.8) gave the cleanest tissue/background separation and was adopted. The stain-labelling threshold is the most delicate: brown ratios are strongly bimodal, and the value 0.02 sits in the valley between the dominant background mode and the positive distribution, on the plateau of the retention curve. Under this label about 15% of patches are positive, quantifying the marked $\approx 1:6$ class imbalance that motivates the sampling strategy described next.

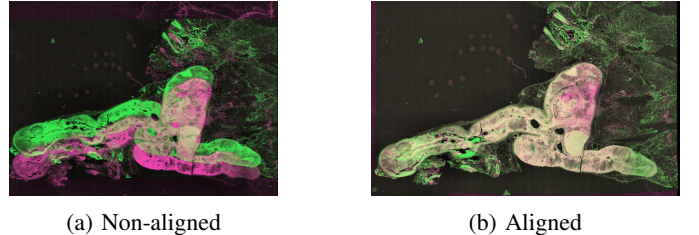


Fig. 1: An example of the alignment process applied to a pair of H&E and IHC WSIs. The left panel shows the original, non-aligned images, while the right panel illustrates the result after applying the VALIS registration framework. The alignment significantly reduces spatial discrepancies, enabling more accurate patch extraction and subsequent model training.

C. Data Augmentation and Sampling Strategy

To mitigate the class imbalance between CYP11B2-positive and negative patches, we implemented a targeted sampling strategy during the training phase. We utilized a weighted random sampler to dynamically adjust the sampling probability, achieving a final training batch distribution of 30% positive and 70% negative patches. This specific 30/70 ratio was selected as an empirical optimal threshold; increasing the proportion of positive patches further risked severe overfitting on the minority class, while this balance sufficiently prevented the model from collapsing into predicting only negative tissue. Alongside the weighted sampling, a comprehensive data augmentation pipeline was applied to all patches to improve generalizability. To preserve the exact spatial correspondence between the H&E inputs and the IHC targets, geometric

transformations were applied simultaneously to both image modalities. The spatial augmentations included:

- **Cropping:** the random 1024×1024 crop and bicubic downscaling to 512×512 described in Section III-B.
- **D4 Symmetry:** Patches were subjected to horizontal flips, vertical flips, and 90-degree rotations.
- **Elastic Deformation:** A conservative elastic transformation ($\alpha=30$, $\sigma=10$) was applied to introduce realistic tissue deformations without disrupting precomputed feature extraction.

Following spatial alignment, photometric augmentations were applied independently to each modality to reflect their specific physical properties. The independent pixel-level transformations were configured as follows:

- **H&E Modality:** Images received color jittering, modifying brightness, contrast, saturation, and hue. This step is critical for simulating the natural staining variability and artifacts encountered across different scanners and laboratory protocols.
- **IHC Modality:** Hue and saturation augmentations were strictly excluded from the IHC patches, because the intensity of the DAB brown chromogen directly encodes the CYP11B2 expression level acting as the ground truth. Instead, the IHC images were only subjected to scanner-realistic artifacts, such as Gaussian noise, Gaussian blur, and minor brightness or contrast adjustments.

D. Model Architectures

We implemented and evaluated four architectures for virtual staining: Multi Channel Correlation Learning (MCCL), UNIStainNet, SwinNet and HistDiST.

1) *Multi Channel Correlation Learning (MCCL):* Multi Channel Correlation Learning (MCCL) [1] was implemented as the primary GAN-based framework for H&E-to-CYP11B2 virtual staining. MCCL explicitly models the biological significance of stain channels by decomposing images into hematoxylin (nuclear morphology), eosin (stromal structure), and DAB (CYP11B2 expression). The architecture employs a CUT-like image translation backbone with a ResNet generator mapping 512×512 H&E patches to synthetic RGB IHC, regulated by a 70×70 PatchGAN discriminator using an LSGAN objective for stable adversarial training.

A key feature of MCCL is its channel-aware supervision. Generated and real IHC patches undergo differentiable HED decomposition, and their respective channels are embedded using a frozen Prov-GigaPath tile encoder [8]. The generator is optimized using a composite objective function that combines several regularizers:

- **Adversarial and Contrastive terms:** Standard LSGAN adversarial loss and PatchNCE contrastive loss to enforce realistic textures and spatial content consistency.
- **Structural Preservation:** An SSIM loss to maintain structural similarity between generated and real IHC patches.
- **Channel-Aware Alignment:** A Channel Knowledge Distillation (CKD) loss to align the GigaPath embeddings of the generated and real IHC channels, and a Multi-Channel

Correlation (MCC) loss to preserve the morphological relationship between the source H&E eosin features and the target IHC channels.

- **DAB Concentration:** To account for the sparsity of CYP11B2 expression, an auxiliary DAB concentration loss applies a weighted Smooth L1 penalty to the generated DAB map, explicitly penalizing under-staining.

The final MCCL generator loss is

$$\mathcal{L}_{\text{MCCL}} = \lambda_{\text{adv}} \mathcal{L}_{\text{LSGAN}} + \lambda_{\text{NCE}} \mathcal{L}_{\text{PatchNCE}} + \lambda_s \mathcal{L}_{\text{SSIM}} + \lambda_{\text{CKD}} \mathcal{L}_{\text{CKD}} + \lambda_{\text{MCC}} \mathcal{L}_{\text{MCC}} + \lambda_{\text{DAB}} \mathcal{L}_{\text{DAB}}^{(\alpha)} \quad (1)$$

This comprehensive objective ensures that the generator balances adversarial realism, structural preservation, contrastive alignment, and DAB-specific reconstruction.

2) *UNIStainNet:* UNIStainNet [2] is a SPADE-UNet [9] GAN whose distinctive trait is that the generator is conditioned *directly* on dense spatial features from a frozen pathology foundation model, rather than through auxiliary losses or externally-derived masks. Generation is a single forward pass, making it lightweight at inference ($\approx 41\text{M}$ trainable parameters, excluding the backbone). The conditioning signal is produced by the UNI [6] ViT-L/16: instead of feeding the whole 512×512 image as a single coarse token sequence, we tile it into a 4×4 grid of sub-crops processed independently and reassemble the patch tokens into a dense map $U \in \mathbb{R}^{32 \times 32 \times 1024}$, where every location is backed by genuine foundation-model features. The overall architecture is shown in Fig. 3.

The generator is a convolutional encoder-decoder with skip connections and a self-attention bottleneck. At each decoder stage a SPADE block injects the conditioning through *dual modulation*:

$$h' = (\gamma^{\text{UNI}} + \gamma^{\text{cls}}) \odot \hat{h} + (\beta^{\text{UNI}} + \beta^{\text{cls}}), \quad (2)$$

where $\hat{h}$ is the instance-normalized activation, the spatially-varying $\gamma^{\text{UNI}}, \beta^{\text{UNI}}$ are convolved from the UNI map (pixel-wise tissue guidance), and the channel-wise $\gamma^{\text{cls}}, \beta^{\text{cls}}$ come from a FiLM-modulated conditioning embedding. The SPADE branch is zero-initialized (ControlNet-style [15]) to stabilize early training. Since the output is pixel-aligned with its H&E input but *not* with the consecutive-section IHC target, every loss is misalignment-aware: perceptual (LPIPS) and L_1 terms act at reduced resolution, a Sobel edge loss enforces structure on the aligned H&E $\rightarrow$ generated axis, adversarial training uses an *unconditional* multi-scale PatchGAN with R_1 penalty and feature matching, and a DAB intensity loss supervises CYP11B2 quantification. The total generator objective combines these terms as

$$\mathcal{L}_G = \mathcal{L}_{\text{percept}} + \lambda_{L_1} \mathcal{L}_{L_1} + \lambda_{\text{edge}} \mathcal{L}_{\text{edge}} + \mathcal{L}_{\text{adv}} + \lambda_{\text{FM}} \mathcal{L}_{\text{FM}} + \lambda_{\text{DAB}} \mathcal{L}_{\text{DAB}}, \quad (3)$$

with $\lambda_{L_1} = 1.0$, $\lambda_{\text{edge}} = 0.5$, $\lambda_{\text{FM}} = 10.0$ and $\lambda_{\text{DAB}} = 0.2$, following the original formulation.

Our setting departs from the original work in two respects. First, UNIStainNet was designed as a *unified multi-stain* generator whose FiLM token selects among several IHC markers; as we target the single CYP11B2 marker, the token instead

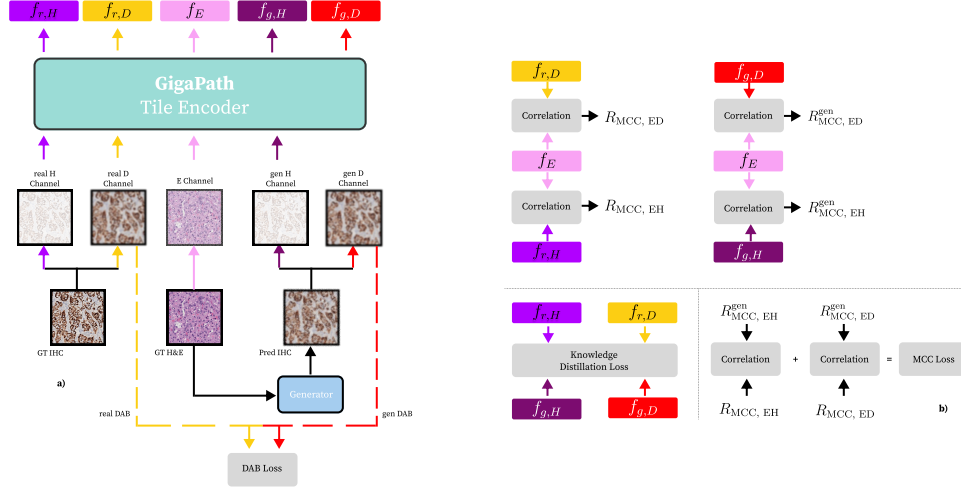


Fig. 2: **MCCL architecture**: the generator translates H&E into synthetic CYP11B2 IHC, while HED decomposition and a frozen Prov-GigaPath encoder provide channel-aware CKD and MCC supervision. The added DAB concentration loss directly constrains the sparse brown immunohistochemical signal.

encodes the patch *positivity* (*unstained*, *stained*, or a *null* class for classifier-free guidance). Second, among the configurations we evaluated, the one we report (Section IV-B) retains a *conditional* discriminator and omits the spatial L_1 , edge, and feature-matching terms of Eq. 3 (with $\lambda_{\text{DAB}} = 0.1$), rather than the unconditional, fully paper-aligned design described above. Although the original paper argues against a conditional discriminator on misaligned data, we found this configuration achieves substantially higher recall on densely CYP11B2-positive tissue in our cohort, which we prioritize over the marginal patch-level image-quality gains the paper-faithful, unconditional-discriminator design would otherwise provide.

3) *SwinNet*: SwinNet is a deterministic paired image-to-image architecture designed to translate 512×512 H&E RGB patches, normalized in $[-1, 1]$, into virtual CYP11B2-IHC RGB patches. The model employs a dual-encoder structure that integrates two complementary feature streams:

- **Semantic Branch**: A pre-trained CTransPath/Swin encoder [10], [16] receives an internally resized 224×224 input normalized with the pretrained mean and standard deviation. The encoder provides multiscale pathology-aware context, with the final stages available for fine-tuning.
- **Structural Branch**: A full-resolution U-Net CNN encoder [17] with 64 base channels preserves local morphology, sharp tissue boundaries, and nuclear details through skip features at 512, 256, 128, 64, and 32 pixel scales (five resolution levels down to a 32×32 bottleneck).

These dual representations are fused by a structure-guided U-Net decoder [18], [19]. Structural skip connections are combined with projected CTransPath context at the bottleneck and at the 64, 128, and 256 decoder stages. Multiscale context gates and a final gated context fusion at 512×512 control how semantic information is injected into the full-resolution decoder. The main output is a three-channel virtual IHC image produced with a tanh activation.

SwinNet also contains an auxiliary DAB head that produces a single-channel DAB logit/probability map at 512×512 . This map can be reinjected into the decoder representation through DAB-guided RGB fusion before the final RGB head. A patch-level DAB presence head estimates whether the patch contains positive CYP11B2 signal and can gate the auxiliary DAB map, reducing spurious activation in globally negative patches.

SwinNet is trained in two phases to stabilize the deterministic mapping. The first phase uses a *morphology-preserving objective*, combining Charbonnier reconstruction, structural similarity (SSIM), hematoxylin edge, hematoxylin MS-SSIM, and deep hematoxylin edge constraints:

$$\mathcal{L}_{\text{phase1}} = \mathcal{L}_{\text{Charb}} + \mathcal{L}_{\text{SSIM}} + \mathcal{L}_{\text{hema}}^{\text{edge}} + \mathcal{L}_{\text{hema}}^{\text{MSSIM}} + \mathcal{L}_{\text{hema}}^{\text{deep}}. \quad (4)$$

The second phase introduces a *DAB-aware refinement objective*: the morphology anchor from Phase 1 is retained, but the deep hematoxylin edge term $\mathcal{L}_{\text{hema}}^{\text{deep}}$ is disabled to allow looser structural constraints during DAB-focused fine-tuning. DAB-guided RGB fusion is activated, and DAB localization pressure is split between RGB-derived losses (positive-intensity, false-positive, and negative-patch) and the auxiliary DAB head (focal BCE, Focal-Tversky, area-control, false-positive, and negative-patch):

$$\begin{aligned} \mathcal{L}_{\text{phase2}} = & (\mathcal{L}_{\text{phase1}} - \mathcal{L}_{\text{hema}}^{\text{deep}}) \\ & + \mathcal{L}_{\text{pos}} + \mathcal{L}_{\text{FP}}^{\text{RGB}} + \mathcal{L}_{\text{neg}}^{\text{RGB}} \\ & + \mathcal{L}_{\text{fBCE}} + \mathcal{L}_{\text{FT}} + \mathcal{L}_{\text{area}} + \mathcal{L}_{\text{FP}}^{\text{DAB}} + \mathcal{L}_{\text{neg}}^{\text{DAB}}. \end{aligned} \quad (5)$$

4) *HistDiST*: HistDiST [3] is a Latent Diffusion Model (LDM) framework specifically tailored for high-fidelity H&E-to-IHC stain translation. At its core, the architecture leverages the powerful pre-trained U-Net from Stable Diffusion [20], adapting its generative capabilities to the medical domain. Moving away from adversarial training, HistDiST frames virtual staining as a conditional generative process. To ensure pathological accuracy, it employs a dual-conditioning strategy:

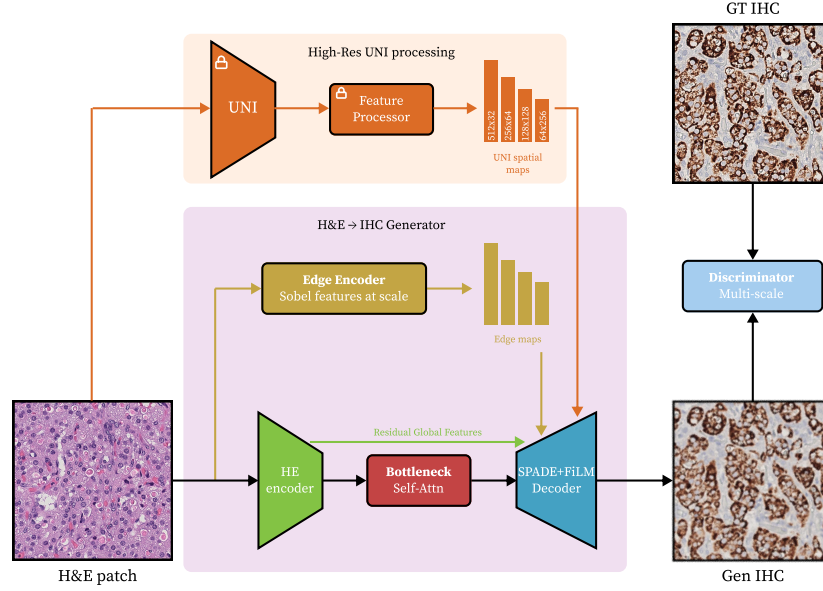


Fig. 3: **UNISTainNet architecture.** The H&E image is split into a 4×4 grid of sub-crops and processed by a frozen UNI ViT-L/16 to produce a dense $32 \times 32 \times 1024$ spatial map. A CNN encoder compresses the H&E input through a self-attention bottleneck; the SPADE+FiLM decoder receives the UNI spatial maps via SPADE modulation (spatially-varying γ_s, β_s) and the patch-positivity embedding via FiLM (channel-wise γ_c, β_c), with encoder skip connections preserving fine-grained detail. An unconditional multi-scale discriminator provides adversarial training.

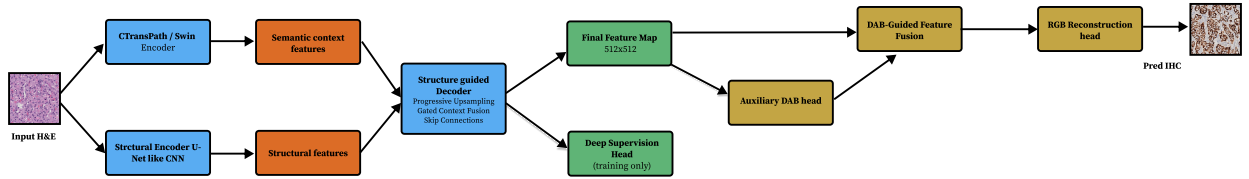


Fig. 4: **SwinNet architecture:** a pathology-pretrained CTransPath/Swin branch extracts multiscale semantic features, while a full-resolution CNN encoder preserves local H&E morphology. A structure-guided decoder progressively fuses both feature hierarchies through gated context fusion. A DAB branch produces an internal probability map that can be reinjected before RGB CYP11B2-IHC reconstruction; a patch-level presence head can gate the DAB map to suppress globally negative patches.

the denoising network is simultaneously guided by morphological embeddings extracted via a foundation model (Phikon) and VAE-encoded H&E representations. Furthermore, the model addresses common diffusion brightness biases by utilizing v -prediction and a rescaled noise schedule. During inference, HistDiST leverages DDIM inversion to strictly anchor the tissue’s structural layout, coupled with an η -cosine noise schedule that introduces controlled stochasticity to capture diverse, biologically plausible molecular expressions without degrading the underlying morphology.

The v -prediction parameterisation [21] defines the network target as a signal-weighted combination of the clean latent x_0 and the noise ϵ :

$$v_t = \sqrt{\alpha_t} \epsilon - \sqrt{1 - \alpha_t} x_0 \quad (6)$$

The latent denoising process is then expressed as a function of the predicted v_t :

$$z_{t-1} = \sqrt{\alpha_{t-1}} \left(\frac{x_t - \sqrt{1 - \alpha_t} v_t}{\sqrt{\alpha_t}} \right) + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} v_t + \sigma_t \eta_t \epsilon \quad (7)$$

To tailor the diffusion training process to the realities of imperfect clinical data, we adapted the standard objective function. While diffusion models predominantly rely on Mean Squared Error (MSE) for the denoising objective, MSE heavily penalizes the intrinsic spatial misalignments frequently present in real-world H&E and IHC pairs. Inspired by recent robust training strategies, we evaluated two distinct optimization regimes: the standard MSE v -loss and a hybrid loss function combining MSE and Mean Absolute Error (MAE). These can be described by a generalized formulation:

$$\mathcal{L} = \lambda_{L_2} \cdot \|v_t - \tilde{v}_t\|^2 + \lambda_{L_1} \cdot \|v_t - \tilde{v}_t\| \quad (8)$$

When configured as a composite objective, this approach maintains high-frequency structural fidelity through the MSE component, while the MAE term mitigates the penalty for minor, unavoidable morphological shifts, preventing the model from over-correcting unaligned artifacts.

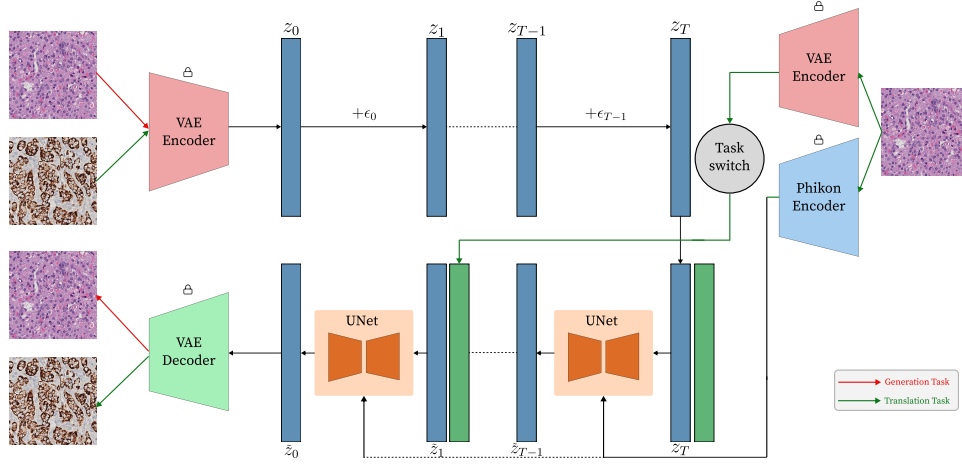


Fig. 5: **HistDiST framework architecture**: the top row depicts the forward diffusion process within the latent space of a frozen VAE ($z_0 \rightarrow z_T$). The bottom row illustrates the iterative backward denoising process driven by a UNet ($\tilde{z}_T \rightarrow \tilde{z}_0$). The architecture incorporates a *Task switch* to dynamically alternate between the *Generation Task* (red arrows) and the *Translation Task* (green arrows), toggling the spatial conditioning channel: the H&E latent is concatenated to the noisy input for translation, while it is replaced by a zero tensor for generation. Semantic features extracted by the frozen Phikon pathology foundation model are injected via cross-attention in both tasks, providing morphological context to the denoising process.

E. Evaluation Metrics

The generated virtual IHC images were evaluated using a comprehensive multi-level pipeline designed to assess pixel-level fidelity, structural morphology, clinical DAB localization, and overall distribution realism. Metrics were computed on the validation split for model selection, and the final reported performance was evaluated on the held-out test set.

1) *Patch and Distribution-Level Evaluation*: We employed standard image reconstruction metrics to evaluate paired fidelity, including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), Structural Correlation Metric (SCM), and Pearson correlation in both RGB and HED-DAB color spaces. To assess the generated dataset’s overall realism and diversity, we computed the Fréchet Inception Distance (FID) and its pathology-specific variant (PathFID) using pre-trained foundation models (e.g., UNI, Giga-Path). Mode collapse and distribution diversity were further monitored via feature-space metrics such as the effective-rank ratio and case-balanced DAB distribution distances.

2) *Region and Morphology-Level Evaluation*: Beyond patch-wise comparisons, we evaluated the clinical utility of the models by reconstructing the slide-level masks to assess DAB localization. We quantified the overlap of predicted CYP11B2-positive regions against the ground truth using the Dice coefficient and Intersection over Union (IoU), supplemented by standard pixel-level classification metrics (precision, recall, specificity, and false-positive rate).

To ensure morphological accuracy, boundary agreement was measured via the Boundary F1 score. Furthermore, object-level detection of positive nodules was evaluated using centroid distance matching to compute an object-level F1 score. The structural plausibility of the correctly identified nodules was validated by comparing key geometric properties, including area, perimeter, eccentricity, solidity, and circularity.

3) *Model Selection Protocol*: The best checkpoint for each architecture was selected based on a multi-criterion strategy on the validation set, ensuring all candidates were evaluated under consistent geometry and preprocessing policies. Priority was given to metrics directly reflecting clinical utility and structural accuracy: DAB Pearson correlation, region-level Dice, nodule F1 score, PathFID on stained regions, and strict guardrail metrics controlling false-positive DAB activation. The held-out test split was accessed exclusively for the final evaluation.

IV. RESULTS AND DISCUSSION

In this section, we present a comprehensive evaluation of the virtual staining models, first focusing on individual models and architectures, and then comparing their performance selecting the best-performing checkpoint for each. We conducted our experiments on a cluster of NVIDIA A40 GPUs, each equipped with 48 GB of VRAM.

A. MCCL

MCCL model selection was performed only on the validation split. To ensure clinical utility, we prioritized DAB-aware metrics (e.g., stained Pearson-DAB, DAB JS, nodule F_1 , and region Dice) over purely perceptual scores. The optimization followed a sequential tuning strategy: selecting the adversarial objective, calibrating DAB-specific losses, and refining hyperparameters. Detailed ablation results are reported in Appendix A.

Our preliminary experiments compared Hinge and LSGAN adversarial objectives. While Hinge models achieved nominally higher Dice scores, they suffered from severe DAB hallucinations (unstained false-positive rate > 0.93). Consequently, LSGAN was selected as a safer baseline. We then introduced explicit DAB supervision. To balance region overlap against false positives, we empirically set $\lambda_{\text{DAB}} = 2.5$ and the DAB

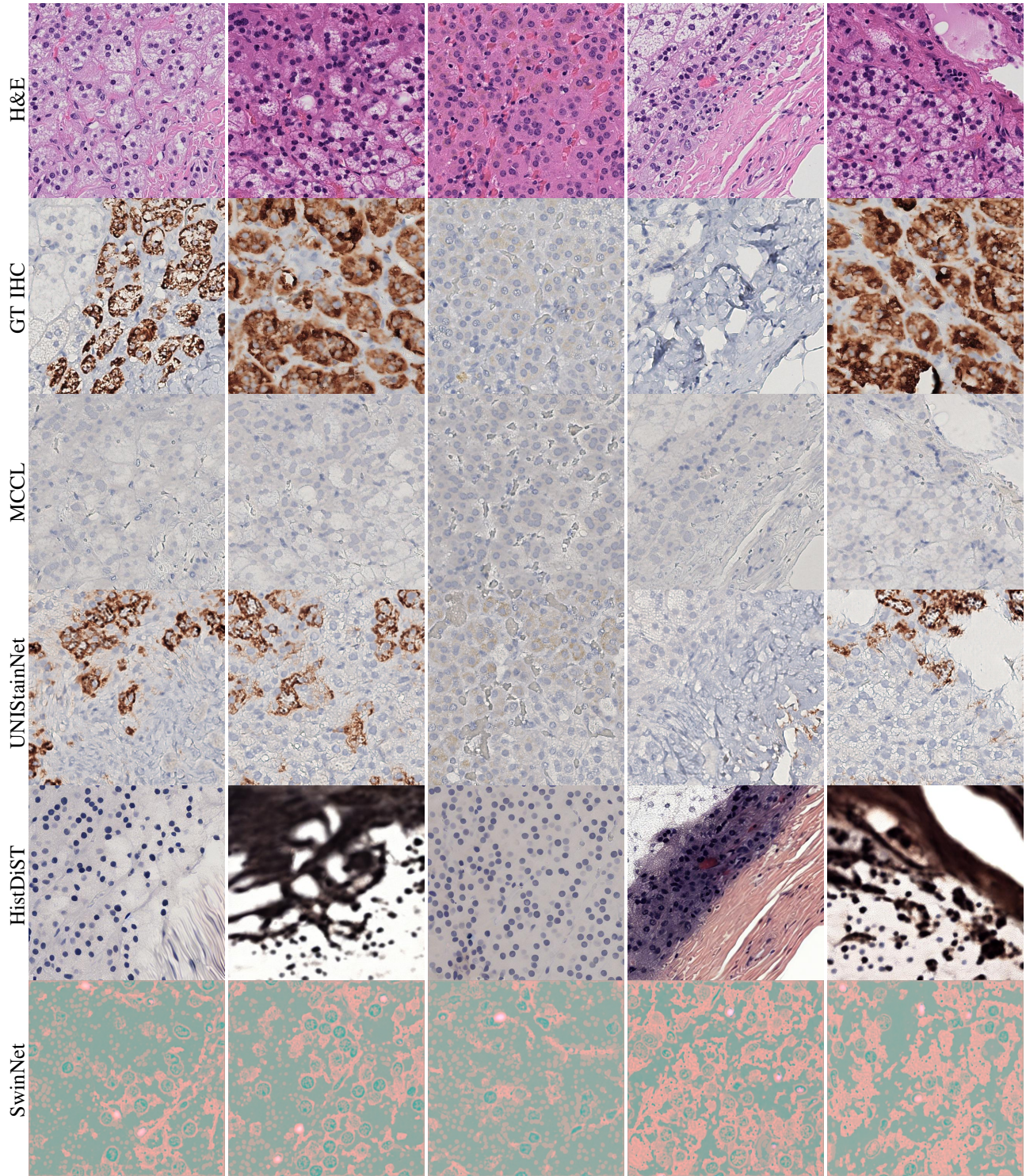


Fig. 6: Qualitative comparison of patch translation across architectures.

TABLE I: Patch metrics comparison of the best-performing checkpoints for each architecture.

Model	MAE ↓	PSNR ↑	RMSE ↓	SCM ↑	SSIM ↑	FID ↓
MCCL	0.1187 ± 0.0435	16.8405 ± 2.5139	0.1511 ± 0.0563	0.0420 ± 0.0231	0.1459 ± 0.0523	96.9332
UNIStainNet	0.1281 ± 0.0448	16.143 ± 2.429	0.1631 ± 0.0571	0.0331 ± 0.0389	0.1108 ± 0.0515	60.2103
SwinNet	0.1499 ± 0.0306	15.0726 ± 1.6018	0.1796 ± 0.0377	0.1264 ± 0.0690	0.2887 ± 0.0930	337.8194
HistDiST	0.3466 ± 0.0618	7.9010 ± 1.4703	0.4084 ± 0.0692	0.0468 ± 0.0287	0.0487 ± 0.0282	293.7585

TABLE II: Region metrics comparison of the best-performing checkpoints for each architecture.

Architecture	Dice ↑	IoU ↑	Boundary F_1 ↑	DAB precision ↑	DAB Recall ↑
MCCL	0.1102 ± 0.0684	0.0593 ± 0.0391	0.1712 ± 0.0384	0.1053 ± 0.1051	0.1721 ± 0.0124
UNIStainNet	0.3667 ± 0.0348	0.2249 ± 0.0258	0.5599 ± 0.0348	0.3461 ± 0.0368	0.3906 ± 0.0383
SwinNet	0.2178 ± 0.1380	0.1269 ± 0.0907	0.2266 ± 0.0875	0.1278 ± 0.0921	0.9776 ± 0.0210
HistDiST	0.2926 ± 0.0088	0.1718 ± 0.0291	0.1394 ± 0.0088	0.2438 ± 0.0541	0.3712 ± 0.0185

positive-pixel weight $\alpha = 3.5$, alongside an Adam momentum of $\beta_1 = 0.1$. This configuration successfully recovered stained DAB and region area, albeit at a minor specificity cost.

Upon final evaluation on the held-out test split (Tables I and II), the selected MCCL model demonstrated a conservative, low-dynamic-range DAB generation. Visual inspection clarified that the false positives were usually not strong visible brown hallucinations; they were mostly diffuse low-level DAB-channel activations under the fixed threshold. The final MCCL model is therefore a valid endpoint of the completed sweep, but its weak region and nodule recovery prevents it from being a clinically reliable virtual CYP11B2 stain. Future MCCL work should move beyond scalar loss sweeps toward explicitly DAB-localized supervision, stronger intensity calibration, and better handling of H&E-IHC misregistration.

B. UNIStainNet

We trained UNIStainNet for 50 epochs (batch size 4, `bfloat16`) with a Two Time-scale Update Rule and the weighted 30/70 positive/negative sampling of Section III. We explored several configurations — varying the discriminator (conditional vs. unconditional) and the spatial/feature-matching loss terms — and report the conditional-discriminator configuration described in Section III-D2, selected for its substantially higher recall on densely CYP11B2-positive tissue, which we prioritize over the marginal patch-level image-quality gains of the paper-faithful, unconditional-discriminator design.

The pixel-level scores are modest (PSNR 16.14, SSIM 0.111, SCM 0.033, Table I). This is expected rather than alarming: these metrics compare output pixels against a consecutive-section target that is never perfectly aligned, so they systematically under-reward biologically plausible predictions. The region-level, DAB-centric measures (Table II) are more informative: the model reaches a Dice of 0.367 and an IoU of 0.225 against the reference DAB mask, with a pixel recall of 0.391, confirmed by qualitative inspection (Fig. 6) to correctly localize densely positive nodules in most samples. We note that this reference mask is itself obtained by automatic color deconvolution and Otsu thresholding of the real IHC, not

by expert annotation, so these scores quantify agreement with an automatically-extracted proxy rather than a clinical ground truth, and that coverage is low ($\approx 4.9\%$ of the WSI canvas, 3 test slides), so these region scores are indicative only. We note that at inference time the positivity token (*stained/unstained*) was provided as a conditioning input, consistent with the original paper’s design; a fully blind evaluation using the null class would represent a more stringent test of the model’s ability to localize aldosterone-producing regions from morphology alone.

C. SwinNet

SwinNet was trained in two consecutive phases to decouple morphology learning from DAB localization, motivated by the deterministic nature of the model and the severe class imbalance. Unlike the other architectures, a uniform sampling ratio and no photometric augmentation on the IHC target were used throughout, to reduce confounding factors during the morphology-first phase.

In phase 1 (50 epochs, $\text{lr } 10^{-4}$, four warmup epochs), only morphology-preserving losses were active: Charbonnier/L1, SSIM, hematoxylin edge, hematoxylin MS-SSIM, and deep hematoxylin edge with weights 0.05, 0.25, 0.20, 0.20, and 0.05. In phase 2 (15 epochs, $\text{lr } 3 \times 10^{-5}$), the morphology checkpoint was fine-tuned with DAB supervision. The morphology anchor was retained, DAB-guided RGB fusion was activated (initial scale 0.25), and DAB localization pressure was split between RGB-derived losses (positive-intensity, false-positive, and negative-patch weights: 0.005, 0.15, 0.25) and the auxiliary DAB head (focal BCE, Focal-Tversky, area-control, false-positive, and negative-patch weights: 0.25, 0.05, 0.03, 0.75, 1.0). Patch-level DAB presence objectives (weight 1.0, area threshold 0.01) further suppressed spurious activation in globally negative patches.

As shown in Tables I and II, SwinNet achieved moderate patch-level quality but failed at DAB localization. The near-perfect DAB recall (0.978) was entirely driven by severe over-prediction: the model assigns DAB positivity to most tissue regions, yielding no meaningful specificity. Nodule-level analysis confirmed that the generated DAB signal did not

correspond to biologically meaningful aldosterone-producing structures.

This outcome is consistent with regression to the mean: under H&E/IHC misalignment, sparse DAB positivity, and ambiguous morphology-to-DAB correspondence, reconstruction losses favor diffuse stain activation. The uniform sampling strategy, while justified in phase 1, likely amplified this failure in phase 2 by underexposing the model to DAB-positive patches. Increasing DAB supervision pressure did not resolve the issue, instead producing optimization shortcuts ranging from strong DAB underprediction to near all-positive activation. These results indicate that deterministic pixel-wise reconstruction is insufficient for sparse, spatially precise stain translation; stochastic approaches with explicit DAB-aware constraints may be better suited to this task.

D. HistDiST

We conducted two distinct experiments utilizing the HistDiST architecture. While largely adhering to the original design, we adapted the training procedure to accommodate our hardware configuration and dataset size. Specifically, HistDiST was trained for 500 epochs using a data-parallel setup across 4 GPUs. To account for the increased global batch size relative to the original implementation—which utilized a single GPU for 1000 epochs on a smaller dataset—we adjusted the learning rate according to the linear scaling rule. Under this adjusted regime, we evaluated two different configurations for the loss function. First, referring to Eq. (III-D4), we set $\lambda_{L_2} = 1$ and $\lambda_{L_1} = 0$, thereby utilizing exclusively the MSE component. In the second configuration, inspired by the work in [13], we adopted a composite loss function by setting $\lambda_{L_2} = 0.7$ and $\lambda_{L_1} = 0.3$, which should punish the model less for bigger spatial misalignments.

Our findings indicate that diffusion architectures exhibit significant limitations when trained on the provided dataset. This drawback primarily stems from numerous instances of spatial misalignment between the H&E and IHC images, which often display completely different visual characteristics. Such structural discrepancies heavily penalize a Latent Diffusion approach, as the model attempts to denoise a latent representation conditioned on geometric and morphological features extracted from the misaligned H&E source.

Consequently, the performance of this architecture was not competitive with GAN-based approaches. GANs proved to be substantially more robust to spatial misalignments, likely because they operate directly on the entire image space rather than relying strictly on a rigid latent representation.

Based on the quantitative results reported in Appendix B (Table VII), the composite loss configuration (MAE+MSE) was selected as the best model for this architecture for the global comparison. As hypothesized, mitigating the impact of spatial pixel shifts allowed this variant to achieve superior performance across key pixel-level metrics such as MAE, PSNR, and RMSE, despite the marginal drop in structural metrics like SCM and SSIM.

Finally, we attempted to integrate an LPIPS loss function by decoding the generated latents during training. However, this

approach introduced a massive computational overhead and consistently led to mode collapse; therefore, these negative results were omitted from the final report.

V. CONCLUSION AND FUTURE WORK

In this work, we explored the capabilities of advanced generative frameworks, including diffusion-based models and GAN based approaches, for the complex task of H&E-to-IHC virtual staining. While these architectures can achieve high visual realism on curated public benchmarks, our investigation underscores that stain transfer remains an inherently formidable challenge, heavily constrained by a fundamental principle: the model is ultimately only as good as the data it is trained on.

A critical open question in the field is whether current generative models are learning a genuine, deterministic mapping of molecular expressions based on tissue morphology, or if they are merely hallucinating biologically plausible IHC signals drawn from the training distribution. This ambiguity becomes a severe limitation when dealing with real-world clinical data. Virtual staining relies on the assumption of strong morphological correspondence between the source and target domains. If the H&E and IHC slides exhibit significant spatial divergence, tissue tearing, or morphological shifts—which are ubiquitous in standard consecutive tissue sectioning—the translation task transitions from highly complex to practically impossible, inevitably yielding suboptimal and unreliable predictions.

Crucially, our experimental observations reveal a stark discrepancy between standard academic benchmarks and real-world clinical data. Widely used public datasets often consist of highly curated, well-aligned image pairs that do not reflect the physical realities of the laboratory. When evaluated on the real-world, routinely prepared clinical slides at our disposal, the inherent misalignment severely degraded the translation quality. The core issue lies not merely in fluctuating quantitative scores, but in the practical, diagnostic utility of the generated images: a visually convincing output is functionally useless if the molecular expression is spatially misplaced.

This practical disconnect highlights a major flaw in how virtual staining models are currently evaluated. Standard metrics (such as FID, SSIM, or feature-based distances) assess global image quality and general structural similarity, but they consistently fail to quantify the true clinical objective. Specifically, these metrics do not account for the direct spatial coincidence between the predicted and ground-truth DAB expression masks. A model might generate a highly realistic image with excellent traditional scores while completely failing to place the DAB positive stain on the correct biological targets.

Future Work. Consequently, future research must pivot from purely architectural enhancements toward addressing these fundamental data-centric and evaluative limitations. Moving forward, it is imperative to develop generative frameworks capable of handling weakly aligned data without relying on pixel-perfect supervision. Furthermore, the community must establish robust, domain-specific evaluation metrics that

explicitly compute the overlap of DAB expression areas to ensure functional and diagnostic accuracy rather than mere visual realism. Finally, integrating rigorous uncertainty quantification mechanisms will be essential to distinguish between reliable morphological translation and statistical hallucination, truly bridging the gap between computational benchmarks and the practical deployment of virtual staining in diagnostic pathology.

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APPENDIX A

MCCL ABLATION STUDIES

This appendix provides the detailed quantitative results from the sequential hyperparameter tuning of the MCCL architecture. Table III summarizes the one-factor-at-a-time ablations used to calibrate the adversarial strength, DAB concentration penalty, positive-pixel weighting, and optimizer momentum.

Subsequent focused sweeps were conducted for the Adam β_1 momentum (Table IV), the DAB weight λ_{DAB} (Table V), and the positive pixel weight α (Table VI).

APPENDIX B

HISTDIST ABLATION STUDIES

This appendix provides the quantitative comparison between the two loss function configurations evaluated for the HistDist architecture.

TABLE III: Detailed one-factor-at-a-time ablations around the LSGAN-DAB baseline. “Winner” indicates the setting carried forward for that axis, not necessarily a final global optimum.

Block / goal	Run or setting	Pearson-DAB $\uparrow$	Unstained FPR $\downarrow$	Dice $\uparrow$	Nodule F_1 $\uparrow$	PathFID $\downarrow$	Decision
Adversarial strength recover sharper DAB	baseline adv 1.0	0.0149	0.0171	0.0544	0.0275	1239.0	winner: keep adv 1.0
	adv 1.5	0.0128	0.8072	0.2256	0.0237	1356.4	rejected: severe FPR
	adv 2.0	0.0194	0.3199	0.1730	0.0715	1304.6	rejected: still unsafe FPR
DAB concentration strength increase brown recovery	baseline DAB 1.0	0.0149	0.0171	0.0544	0.0275	1239.0	conservative reference
	DAB 1.8	0.0175	0.0052	0.0061	0.0115	1454.0	rejected: under-staining
	DAB 2.5	0.0294	0.1621	0.2000	0.0843	1125.6	winner: tune further
DAB positive weighting rebalance sparse positives	baseline $\alpha = 5.0$	0.0149	0.0171	0.0544	0.0275	1239.0	specificity reference
	$\alpha = 2.5$	0.0179	0.0424	0.0581	0.0582	1125.0	winner candidate: tune
	$\alpha = 10.0$	0.0031	0.9609	0.2425	0.0006	1432.1	rejected: saturation
Structural SSIM weight test stronger structure	baseline $\lambda_s = 0.05$	0.0149	0.0171	0.0544	0.0275	1239.0	winner: keep 0.05
	$\lambda_s = 0.15$	0.0063	0.9403	0.2408	0.0018	1411.2	rejected: false positives
	$\lambda_s = 0.30$	0.0130	1.0000	0.2428	0.0000	2709.5	rejected: collapse
Optimizer momentum increase responsiveness	baseline $\beta_1 = 0.5$	0.0149	0.0171	0.0544	0.0275	1239.0	conservative reference
	$\beta_1 = 0.0$	0.0125	0.0000	0.0000	0.0000	2003.7	rejected: no DAB
	$\beta_1 = 0.1$	0.0211	0.2523	0.2013	0.0875	1082.7	winner candidate: tune

TABLE IV: Focused Adam β_1 sweep. The value $\beta_1 = 0.1$ was selected as the most useful sensitivity-oriented setting. $\beta_1 = 0.4$ increased Pearson-DAB and Dice, but its false positives were too high; $\beta_1 = 0.0$ and 0.2 suppressed DAB almost entirely.

β_1	Pearson-DAB $\uparrow$	DAB JS $\downarrow$	Unstained FPR $\downarrow$	Dice $\uparrow$	Nodule F_1 $\uparrow$	PathFID $\downarrow$
0.0	0.0125	1.0000	0.0000	0.0000	0.0000	2003.7
0.1	0.0211	0.6259	0.2523	0.2013	0.0875	1082.7
0.2	0.0154	1.0000	0.0001	0.0001	0.0002	1380.1
0.3	0.0182	0.9530	0.0279	0.0563	0.0413	1131.0
0.4	0.0251	0.7152	0.4339	0.2088	0.0602	1093.2
0.5	0.0149	0.9584	0.0171	0.0544	0.0275	1239.0

TABLE V: Sweep of λ_{DAB} after selecting $\beta_1 = 0.1$. The final choice from this stage was $\lambda_{\text{DAB}} = 2.5$: it strongly reduced false positives relative to $\lambda_{\text{DAB}} = 1.0$, kept non-zero region recovery, and achieved the best stained PathFID.

λ_{DAB}	Pearson-DAB $\uparrow$	DAB JS $\downarrow$	Unstained FPR $\downarrow$	Dice $\uparrow$	Nodule F_1 $\uparrow$	PathFID $\downarrow$	Region FPR $\downarrow$	Coverage $\uparrow$
1.0	0.0211	0.6259	0.2523	0.2013	0.0875	1082.7	0.2721	0.5369
2.2	0.0162	0.9343	0.0639	0.0948	0.0591	1171.1	0.0756	0.0390
2.5	0.0177	0.9635	0.0511	0.0819	0.0637	984.2	0.0566	0.0803
2.8	0.0165	0.9345	0.1355	0.1730	0.0482	1056.8	0.1489	1.0000
3.2	0.0158	1.0000	0.0017	0.0013	0.0024	1213.6	0.0007	0.0000

TABLE VI: Stage 4C positive-pixel weight sweep after the selected Stage 4B configuration. The selected setting gave the strongest stained DAB recovery and region overlap, while the baseline remained the cleaner specificity and realism reference.

α	Pearson-DAB $\uparrow$	DAB JS $\downarrow$	Unstained FPR $\downarrow$	Dice $\uparrow$	Nodule F_1 $\uparrow$	Stained PathFID $\downarrow$	Overall PathFID $\downarrow$	Region FPR $\downarrow$	Stained cov. $\uparrow$
0.5	0.0177	0.9635	0.0511	0.0819	0.0637	984.2	503.9	0.0566	0.0803
2.5	0.0189	1.0000	0.0023	0.0030	0.0043	1529.1	1130.2	0.0020	0.0000
3.5	0.0229	0.8603	0.1456	0.1511	0.0620	1090.2	559.2	0.1717	0.6707
7.5	0.0178	0.9805	0.0339	0.0636	0.0552	1018.0	550.9	0.0422	0.0000

TABLE VII: Metric comparison between the two HistDiST configurations. The composite loss function (MAE+MSE) achieves superior performance in terms of MAE, PSNR, RMSE and FID, while the MSE-only configuration slightly outperforms in SCM and SSIM.

Model	MAE $\downarrow$	PSNR $\uparrow$	RMSE $\downarrow$	SCM $\uparrow$	SSIM $\uparrow$	FID $\downarrow$
MAE+MSE	0.3466 $\pm$ 0.0618	7.9010 $\pm$ 1.4703	0.4084 $\pm$ 0.0692	0.0468 $\pm$ 0.0287	0.0487 $\pm$ 0.0282	293.7585
MSE Only	0.3555 $\pm$ 0.0640	7.7192 $\pm$ 1.4741	0.4171 $\pm$ 0.0714	0.0525 $\pm$ 0.0331	0.0497 $\pm$ 0.0296	301.8802